

Problem Set for 19-May and 22-May

Part I: the Kronecker product, tensor products for matrices

Example Take

• $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}, Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix}$. Now

1. The matrix equation $AX = Y$ coincides the linear system

$$\begin{pmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & a_{11} & 0 & a_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & a_{21} & 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

2. The matrix equation $XB = Y$ coincides linear system

$$\begin{pmatrix} b_{11} & b_{21} & 0 & 0 \\ b_{12} & b_{22} & 0 & 0 \\ 0 & 0 & b_{11} & b_{21} \\ 0 & 0 & b_{12} & b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

3. The matrix equation $AXB = Y$ coincides linear system

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

where we write

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}B^T & a_{12}B^T \\ a_{21}B^T & a_{22}B^T \end{pmatrix} =: A \otimes B^T.$$

Definition (Kronecker product of matrices) For $P \in K^{k \times l}, Q \in K^{m \times n}$, define

$$P \otimes Q = \begin{pmatrix} p_{11}Q & p_{12}Q & \cdots & p_{1l}Q \\ p_{21}Q & p_{22}Q & \cdots & p_{2l}Q \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1}Q & p_{k2}Q & \cdots & p_{kl}Q \end{pmatrix}.$$

Clearly, $P \otimes Q \in K^{km \times ln}$.

Exercise 1 Verify the following statements yourself (**not a homework**):

1. $\otimes$ is an binary operation which is associative;
2. $\otimes$ is bilinear on both sides;
3. $(A \otimes B) \cdot (C \otimes D) = (A \cdot C) \otimes (B \cdot D)$;
4. $(A \otimes B)^T = A^T \otimes B^T$;
5. $\text{rank}(A \otimes B) = \text{rank}(A) \cdot \text{rank}(B)$;

Now we assume that A and B are square matrices:

1. $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$;
2. $\det(A \otimes B) = (\det A)^{\dim A} (\det B)^{\dim B}$;
3. $\text{tr}(A \otimes B) = \text{tr}(A) \cdot \text{tr}(B)$;
4. Let (λ, v) and (μ, u) be eigenpairs of A and B , respectively. Then $A \otimes B$ has $(\lambda\mu, v \otimes u)$ as an eigenpair;

Exercise 2 Let J_A and J_B denote the Jordan form of A and B respectively (all matrices are in $\mathbb{F}^{n \times n}$).

1. Write down the Jordan form of the linear transformation $X \mapsto AXA^T$;
2. Write down the Jordan form of the linear transformation $X \mapsto AXA$;
3. Write down the Jordan form of the linear transformation $X \mapsto AX - XA$;
4. Write down the Jordan form of the linear transformation $X \mapsto AX - XA^T$;
5. Write down the Jordan form of the linear transformation $X \mapsto AXB$.

同一解答 1, 2. 与 5. 记 $A \oplus B = \begin{pmatrix} A & O \\ O & B \end{pmatrix}$ 是分块形式, 则

$$\left(\bigoplus_{1 \leq i \leq n} A_i \right) \otimes \left(\bigoplus_{1 \leq j \leq m} B_j \right) = \bigoplus_{1 \leq i \leq n; 1 \leq j \leq m} A_i \otimes B_j.$$

为求 $J_A \otimes J_A$ 的 Jordan 标准型, 只需考虑 Jordan 块 $J_s(\lambda) \otimes J_t(\mu)$ 的 Jordan 标准型. 这是个纯组合的问题:

1. $\mu\lambda \neq 0$. 此时

$$J_s(\lambda) \otimes J_t(\mu) \sim J_{s+t-1}(\lambda\mu) \oplus J_{s+t-3}(\lambda\mu) \oplus \cdots \oplus J_{|s-t|+1}(\lambda\mu).$$

2. $\mu = 0, \lambda \neq 0$, 则 $J_s(\lambda) \otimes J_t(0)$ 是 s 个分块对角的 $J_t(0)$.

3. $\mu = \lambda = 0$, 则

$$J_s(0) \otimes J_t(0) \sim \left(\bigoplus_{l=1}^{\min(s,t)-1} J_l(0) \oplus J_l(0) \right) \oplus \left(\bigoplus_{k=1}^{|s-t|+1} J_{\min(s,t)}(0) \right).$$

对3. 与4., 求解 $J_A \otimes I - I \otimes J_B^T$ 的 Jordan 标准型即可. 对线性映射

$$X \mapsto J_A X - X J_B,$$

可以将 X 进行分块, 其纵向分割线由 J_A 的分块决定, 横向分割线由 J_B 的分块决定. 此处, X 的每一分块是线性映射的不变子空间. 该映射限制在不变子空间上形如

$$\mathbb{F}^{s \times t} \rightarrow \mathbb{F}^{s \times t}, \quad B \mapsto J_s(\lambda)B - B J_t(\mu).$$

将上述线性映射减去 $\lambda\mu \cdot \text{id}$, 得幂零变换 $B \mapsto J_s(0)B - B J_t(0)$. 由 $J_t(0) \sim (-J_t(0)^T)$, 这一幂零变换对应矩阵 $J_s(0) \otimes I + I \otimes J_t(0)$. 归纳得 Jordan 标准型

$$J_s(0) \otimes J_t(0) \sim J_{s+t-1}(0) \oplus J_{s+t-3}(0) \oplus \cdots \oplus J_{|s-t|+1}(0).$$

Exercise 3 Show that $\dim \ker[X \mapsto (AX - XA^T)] \geq n$, and explain when the equality holds.

这等价于求解全体与 J_A 可交换矩阵构成的线性空间的维数的极小值. 记 $J_A = \bigoplus_{i=1}^s J_i$, 并依照式 $J_A X - X J_A$ 对 X 进行分块. 分块 $X_{i,j}$ 恒为零, 当且仅当 J_i 与 J_j 没有公共特征值. 特别地, 分块 $X_{i,i}$ 可取的值是 Jordan 块的多项式, 若 $X_{i,i} \in \mathbb{F}^{p \times p}$, 则 $X_{i,i}$ 处可取的矩阵构成 p 维线性空间.

综上, 取等当且仅当 J_A 的 Jordan 块的特征值彼此不同 (不论 Jordan 块大小与数量).

Exercise 4 Show that there exists $\{0, 1\}$ -matrices A and B such that $A \otimes B$ and $B \otimes A$ are not similar.

取 $A = \begin{pmatrix} 0 & I \end{pmatrix} \in \mathbb{F}^{2 \times 4}$ 与 $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{F}^{2 \times 1}$. 显然

$$\text{tr}(A \otimes B) = 1 \neq (1 + 1) = \text{tr}(B \otimes A),$$

因此两者不可能相似.

Tip

For square matrices A and B , it is clear that $A \otimes B \sim B \otimes A$. Proof of the statement: By linear isomorphism of $(-)^T$, $[X \mapsto AXB^T]$ and $[Y \mapsto BYA^T]$ stand for the same linear map under different basis. Hence $(A \otimes B) \sim (B^T \otimes A^T)$. Since $A \sim A^T$ and $B \sim B^T$, one has $(A \otimes B) \sim (B \otimes A)$.

Exercise (optional) If you know the adjacency matrices $A(G_i)$ for a simple graph G_i , then discover the graph operations $\boxplus$ and $\boxtimes$ such that

1. $A(G_1) \otimes A(G_2) = A(G_1 \boxtimes G_2)$;
2. $A(G_1) \otimes I + I \otimes A(G_2) = A(G_1 \boxplus G_2)$.

对两种图, 顶点集均是原顶点集的 Cartesian 积.

$G_1 \boxtimes G_2$ 的边集是通过 and-关系定义的, $(u, v) \sim (u', v')$ 当且仅当 $(u \sim u') \wedge (v \sim v')$.

$G_1 \boxplus G_2$ 的边恰好形如 $(u, v) \sim (u, v')$ 或者 $(u, v) \sim (u', v)$, 其中 $u \sim u', v \sim v'$.

Tip

Kronecker 积 $\otimes$ 的本质是指标的运算, 图论语言可以将这一指标运算可视化.

Part II: the tensor product in general, and the use of 1 : 1 correspondence

Important

Throughout, $\sum$ is always a finite sum.

Definition (Tensor product of two finite dimensional vector spaces). Let $\{u_i\}_{1 \leq i \leq n}$ be a basis of U , and $\{v_j\}_{1 \leq j \leq m}$ be a basis of V . Now one can show the 1 : 1 correspondence of the following three types of functions.

1. a bilinear map from U, V to $\mathbb{F}$;
2. a map of the type $f : S \rightarrow \mathbb{F}$ with property P, where
 - $S := \{M \in \mathbb{F}^{n \times m} \mid \text{rank}(M) = 1\}$, the subspace of rank one matrices;
 - P means that, once there is an equality of linear combination in S (e.g., $\sum c_i s_i = 0$ for $s_i \in S$), one has $\sum c_i f(s_i) = 0$.
3. a linear map $\text{Hom}_{\mathbb{F}}(\mathbb{F}^{n \times m}, \mathbb{F})$.

We identify $U \otimes V := \mathbb{F}^{n \times m}$ in 3. by chasing the basis $\{u_i\}_{1 \leq i \leq n}$ and $\{v_j\}_{1 \leq j \leq m}$ from 1. to 3.:

- $U \otimes V$ has a basis $u_i \otimes v_j$, i.e., the $E_{i,j}$ matrices in the space $\mathbb{F}^{n \times m}$;
- An element of $U \otimes V$ is of the form $\sum \lambda_i (u_i \otimes v_i)$, i.e., the sum of $E_{i,j}$'s;
- Say a $x \in U \otimes V$ is an simple tensor, whenever there is $u \in U$ and $v \in V$ such that $x = u \otimes v$, i.e., the rank-one matrix in $\mathbb{F}^{n \times m}$;
- The tensor-rank of $x \in U \otimes V$ is the least number of simple tensors summing up to x , i.e., 使用相抵标准型将矩阵写作秩 1 矩阵的和.

Definition (The definition of tensor product in general). We consider the properties that a hypothetical object $U \otimes V$ would possess:

1. an object of $x \in U \otimes V$ is the finite sum of simple tensors, e.g., $x = \sum u_t \otimes v_t$ for some $u_t \in U$ and $v_t \in V$;
2. there is a canonical bilinear map $\theta_{U,V} : U \& V \rightarrow U \otimes V$ sending the pair (u, v) to a simple tensor $u \otimes v$, which means that
 1. $(u + u') \otimes v$ and $u \otimes v + u' \otimes v$ are the same elements,
 2. $u \otimes (v + v')$ and $u \otimes v + u \otimes v'$ are the same elements,
 3. $\lambda(u \otimes v)$, $(\lambda u) \otimes v$ and $u \otimes (\lambda v)$ are the same elements;
3. any bilinear form $U \& V \rightarrow W$ corresponds to a linear map from $U \otimes V$ to W , which means that
 1. once we have a bilinear map $B : U \& V \rightarrow W$, we have a linear map $U \otimes V \rightarrow W$ determined by the image of simple tensors:

$$u \otimes v \mapsto B(u, v).$$

2. once we have a linear map $\varphi : U \otimes V \rightarrow W$, there is a bilinear map $\varphi \circ \theta_{U,V}$.

Note

In short, the linear space $\text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(V, W))$ (not $\text{Hom}_{\mathbb{F}}(U \times V, W)$) stands for the set of bilinear maps from $U \& V$ to W . The 1 : 1 correspondence means the bijection

$$\text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(V, W)) \rightarrow \text{Hom}_{\mathbb{F}}(U \otimes V, W), \quad B \mapsto B \circ \theta_{U,V}$$

Such an bijective is also linear (verify it your self). Hence, the above is an linear isomorphism.

The adjunction: $\text{Hom}(A, \text{Hom}(B, C)) \simeq \text{Hom}(A \otimes B, C)$.

Definition ($f \otimes f'$). For linear maps $f : U \rightarrow V$ and $f' : U' \rightarrow V'$, the bilinear map

$$U \& U' \rightarrow V \& V' \rightarrow V \otimes V' \quad (u, u') \mapsto f(u) \otimes f'(u')$$

corresponds to a unique linear map $U \otimes U' \rightarrow V \otimes V'$ sending simple tensor $u \otimes u'$ to $f(u) \otimes f'(u')$. We denote

$$f \otimes f' : U \otimes U' \rightarrow V \otimes V', \quad \sum u \otimes u' \mapsto \sum f(u) \otimes f'(u').$$

Example (The magic of 1 : 1 correspondence). We show that

$(g \otimes g') \circ (f \otimes f') = (g \circ f) \otimes (g' \circ f')$ are the same linear maps from $U \otimes U'$ to $W \otimes W'$, where

$$U \xrightarrow{f} V \xrightarrow{g} W, \quad U' \xrightarrow{f'} V' \xrightarrow{g'} W'.$$

Step 1 $(g \otimes g') \circ (f \otimes f') = (g \circ f) \otimes (g' \circ f')$ whenever $(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'} = (g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'}$ are the same linear map, where $\theta_{U,U'} : U \otimes U' \rightarrow U \otimes U'$ is the canonical bilinear map.

Step 2 We consider $(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'}$. By definition of $f \otimes f'$, we have $(f \otimes f') \circ \theta_{U,U'} = \theta_{V,V'} \circ (f \otimes f')$, the same bilinear map from $U \otimes U'$ to $V \otimes V'$. Hence

$$(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'} = (g \otimes g') \circ \theta_{V,V'} \circ (f \otimes f') = \theta_{W,W'} \circ (g \otimes g') \circ (f \otimes f').$$

Step 2' We consider $(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'}$. By definition, one has

$$(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'} = \theta_{W,W'} \circ ((g \circ f) \otimes (g' \circ f')).$$

Step 3 It is clear that $((g \circ f) \otimes (g' \circ f'))$ and $(g \otimes g') \circ (f \otimes f')$ are the same thing from $U \otimes U' \rightarrow W \otimes W'$. This complete the proof.

Exercise 5 Show by 1 : 1 correspondence that, there exists an isomorphism

$$\Phi_{U,V,W} : U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W.$$

sending simple tensor $u \otimes (v \otimes w)$ to $(u \otimes v) \otimes w$.

(optional) Moreover, such Φ is natural, which means that for $f : U_f \rightarrow V_f, g : U_g \rightarrow V_g$ and $h : U_h \rightarrow V_h$, one has

$$((f \otimes g) \otimes h) \circ \Phi_{U_f, U_g, U_h} = \Phi_{V_f, V_g, V_h} \circ (f \otimes (g \otimes h))$$

Part I 先构造 Φ , 即验证 $\lambda x \mapsto \langle \langle x.1, x.2.1 \rangle, x.2.2 \rangle$.

为构造 Φ , 先考虑 Φ 对应的双线性 $B : U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$.

Step 1 证明对固定的 u_0 , 双线性型

$$B(u_0, \langle -, - \rangle) : V \otimes W \rightarrow (U \otimes V) \otimes W, \quad (v, w) \mapsto (u_0 \otimes v) \otimes w$$

是良定义的. 这一双线性型来自复合

$$V \otimes W \xrightarrow{\star} (U \otimes V) \otimes W \xrightarrow{\dagger} (U \otimes V) \otimes W, \quad (v, w) \mapsto (u_0 \otimes v, w) \mapsto (u_0 \otimes v) \otimes w.$$

- $\star$ 的第一分量来自典范双线性 $U \otimes V \rightarrow U \otimes V$ 右输入, 第二分量是恒等. 从而 $\star$ 良定义.
- $\dagger$ 是典范双线性.

这给出了线性映射

$$B(u_0, -) : V \otimes W \rightarrow (U \otimes V) \otimes W, \quad v \otimes w \mapsto (u_0 \otimes v) \otimes w.$$

Step 2 对每一个 u_0 , 会有一个 $V \otimes W \rightarrow (U \otimes V) \otimes W$ 类型的线性映射. 由张量积的双线性, 对应 $u_0 \mapsto B(u_0, -)$ 是线性的. 具体地说,

$$((u_1 + \lambda u_2) \otimes v) \otimes w = (u_1 \otimes v) \otimes w + \lambda((u_2 \otimes v) \otimes w).$$

Step 3 以上两步给出了双线性映射

$$B(-, -) : U \otimes V \otimes W \rightarrow (U \otimes V) \otimes W, \quad (u, v \otimes w) \mapsto (u \otimes v) \otimes w$$

这诱导了上述 $\Phi_{U,V,W}$.

Part II 证明上述构造的 Φ 是同构, 说明方式是构造逆映射.

然后说明 $\Phi_{U,V,W}$ 是同构. 对称地构造

$$\Psi_{U,V,W} : (U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W),$$

简单张量 $(u \otimes v) \otimes w$ 的像是 $u \otimes (v \otimes w)$. 由 $\Psi_{U,V,W} \circ \Phi_{U,V,W}$ 在简单张量上是恒等, 故 $\Psi_{U,V,W} \circ \Phi_{U,V,W}$ 就是恒等映射; 类似地, $\Phi_{U,V,W} \circ \Psi_{U,V,W}$ 也是恒等映射. 这直接构造了 Φ 的逆映射.

Part III 说明自然性, 也就是

$$((f \otimes g) \otimes h) \circ \Phi_{U_f, U_g, U_h} = \Phi_{V_f, V_g, V_h} \circ (f \otimes (g \otimes h))$$

其实这不难, 比较简单张量的像即可.

Exercise 6 Show that for arbitrary linear surjection p , the linear map $\text{id}_U \otimes p$ is also a surjection.

对满射 $p : V \rightarrow W$, 下证明

$$\text{id}_U \otimes p : U \otimes V \rightarrow U \otimes W, \quad \sum u_i \otimes v_i \mapsto \sum u_i \otimes p(v_i).$$

是满射. 由题意, 任意 $U \otimes W$ 中元素 $\sum u_i \otimes w_i$ 均有原像 $\sum u_i \otimes v_i$, 其中 $p(v_i) = w_i$.

Exercise 7 Show that for arbitrary linear injection i , the linear map $\text{id}_U \otimes i$ is also an injection.

对单射 $i : V \rightarrow W$, 下证明

$$\text{id}_U \otimes i : U \otimes V \rightarrow U \otimes W, \quad \sum u_i \otimes v_i \mapsto \sum u_i \otimes i(v_i).$$

是单射. 任取 $\sum u_i \otimes v_i$, 满足 $\sum u_i \otimes i(v_i) = 0$, 下证明 $\sum u_i \otimes v_i = 0$ 即可.

由于张量是简单张量的有限和, 任意元素能写成数量最少的简单张量的和. 不妨设 $\sum_{i=1}^t u_i \otimes v_i$ 如此. 我们断言全体 $\{v_i\}$ 是线性无关组; 若不然, 则可以重新找一组 $\text{span}(\{v_i\})$ 的基 (大小 $s < t$), 使得 $u_i \otimes v_i = \sum_{j=1}^s u'_{i,j} \otimes v'_j$. 因此,

$$\sum_{i=1}^t u_i \otimes v_i = \sum_{j=1}^s \left(\sum_{i=1}^t u'_{i,j} \right) \otimes v'_j$$

是求和数量更少的简单张量的和, 与假定矛盾.

由单射, $\{i(v_i)\}$ 也是线性无关组. 使用延拓公理, 对任意 j , 存在 $\varphi_j : W \rightarrow \mathbb{F}$ 使得 $\varphi_j(i(v_i)) = \delta_{i,j}$. 因此

$$(\text{id}_U \otimes \varphi_j) : \sum u_i \otimes i(v_i) \mapsto u_j.$$

以上将 0 映作 0 , 从而 $u_j = 0$. 由所有 $u_i = 0$, 只能有 $\sum u_i \otimes v_i = 0$.

💡 Tip

You are allowed to use the following axiom: For any vector subspace $S \subseteq L$, any linear map $f : S \rightarrow W$ extends to $\tilde{f} : L \rightarrow W$.